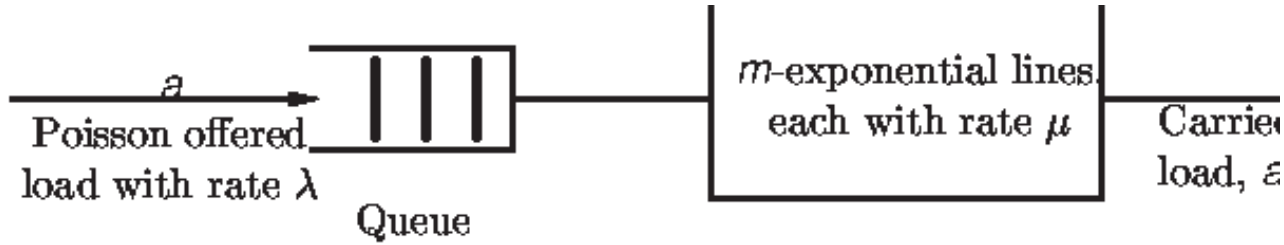
Erlang Delay

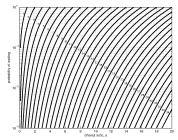
2006-10-16



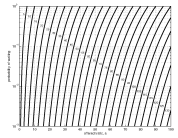
Queueing model of the Erlang delay system.



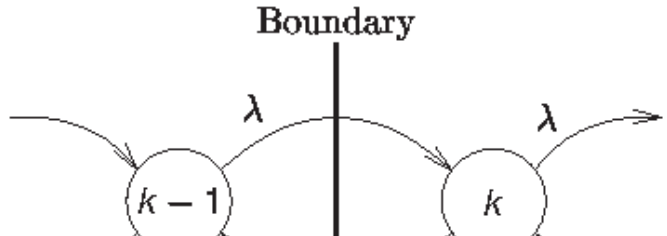
State diagram of an Erlang delay system.



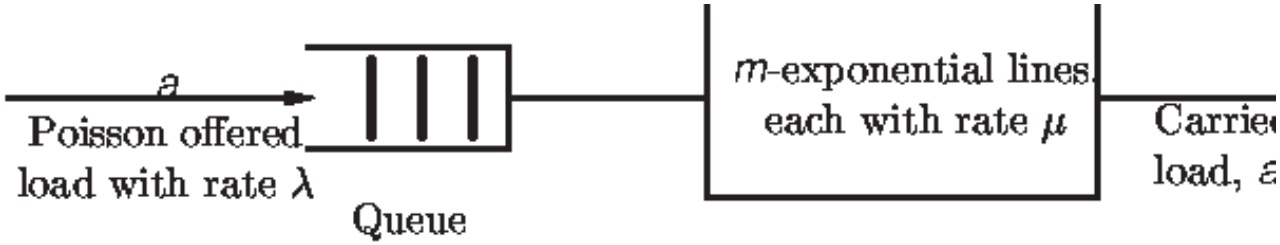
Erlang C curves for offered traffic between 0 and 20.



Erlang C curves for offered traffic between 0 and 100.



Erlang Delay $M/M/m$



erlangDelayDiagram

Erlang Delay $M/M/m$ (continued)

- We have considered the situation were there are as many servers as there are customers.
- More realistic real world situations involve limited numbers of servers.
- For example, call centres.
- The system is known as the *Erlang delay* system. In Kendall's notation it is written $M/M/m$.
- The 'average holding time', $\tau = \frac{1}{\mu} \equiv \bar{X}$, which, for a network involving telephones is the time a call is open.
-

State Diagram



erlangDelay

State Diagram (continued)

- The 'death rate' μ_k has different definitions in different parts of the state space.
- We need to consider detailed balance for each part in turn.

Detailed Balance

- The Erlang delay system may be analysed as a birth-death process with birth rate

$$\lambda_k = \lambda$$

and death rate

$$\mu_k = \begin{cases} k\mu & k = 0, 1, \ldots, m-1 \\ m\mu & k > m \end{cases}$$

}

The Two Situations

- This means we have to consider two situations in detailed balance
- When the system's state, k, is less than or equal to the number of servers.
- When the system's state, k, is greater than the number of servers. We will cover this as an example.

Stationary Distribution

- The overall stationary distribution is then

$$P(k) = \begin{cases} P(0)\frac{(m\rho)^k}{k!}, & k \leq m \\ P(0)\frac{m^m\rho^k}{m!} & k > m \end{cases}$$

$$P(0) = \left[\sum_{k=0}^{m-1}\frac{(m\rho)^k}{k!} + \frac{(m\rho)^m}{m!(1-\rho)}\right]^{-1}.$$

Note that this relies on $\rho < 1$, if $\lambda \geq m\mu$ the system has no stationary distribution.}

Probability of Waiting

- For the $M/M/m$ queue a key quantity of interest is the probability of of a customer queueing, P_Q.
- What is the probability, when the system is at steady state, that a customer will arrive and find all the m servers busy?

$$P(k \geq m) = \sum_{k=m}^{\infty} P(k)$$

$$P(k \geq m) = P(0) \sum_{k=m}^{\infty} \frac{m^m \rho^k}{m!}$$

$$P(k \geq m) = P(0) \frac{(\rho m)^m}{m!} \sum_{k=m}^{\infty} \rho^{k-m},$$

Probability of Waiting contd

- Substituting $\dot{k} = k - m$ in the sum gives

$$P(k \geq m) = P(0) \frac{(\rho m)^m}{m!} \sum_{\dot{k}=1}^{\infty} \rho^{\dot{k}},$$

- Making the usual substitution this can be re-written

$$C(m, a) = P(k \geq m) = P(0) \frac{(\rho m)^m}{m!(1-\rho)}.$$

Erlang C Formula

- This is known as the *Erlang C formula*, or Erlang's second formula after A.K. Erlang. By making use of our definition of a we can rewrite the offered load as

$$a = \frac{\lambda}{\mu} = m\rho$$

and substituting into (11) we have

$$C(m, a) = \frac{P(0)\, a^m}{m!\left(1 - \frac{a}{m}\right)}.$$

Computing the Erlang C Formula

- Can be computed on a computer: but
- some care must be taken when computing it.

Computing the Erlang C Formula (continued)

- when m is large overflow problems will occur for a^m and $m!$.
- Compute the formula in log space,

$$\ln C(m, a) = \ln P(0) + m \ln a - \ln m! - \ln\left(1 - \frac{a}{m}\right).$$

Still problems with $\ln P(0)$ term, but each of the other terms can be computed with appropriate functions in Python (`scipy.special.gammaln`). * When we need to compute the Erlang C curve, for the purposes of exams and exercises, we will use the old fashioned method of curves.

Erlang C Curves

- We have provided you with two sets of curves in your notes.
- In an exam, if they are need curves will be provided with the exam paper.
- Make sure you mark the curves and hand them in with your paper.

Examples

- Examples 2 and 3 at the end of the chapter cover the Erlang C curves.

Average number in System.

- We can also compute the average number of calls in the Erlang Delay system,

$$\bar{N} = \frac{\rho}{1-\rho} C(m, a) + m\rho$$

- These two terms represent the average queue length and the average number of busy lines.
- The average queue length is given by

$$\bar{N}_Q = \frac{\rho}{1-\rho} C(m, a)$$

Average Queueing Time

- Using Little's formula the average waiting time is simply

$$\bar{W} = \frac{\bar{N}_Q}{\lambda} = \frac{C(m, a)}{m\mu(1-\rho)}.$$

- Note that this is the average queueing time for all customers (whether they queued or not).

Queueing Time for those who Queue

- Use the fact that the average queueing time for those who don't queue is zero.
- The average queueing time can be decomposed as

$$\bar{W} = C(m, a)\bar{W}_Q + (1 - C(m, a))\,\bar{W}_{\sim Q}$$

Queueing Time for those who Queue (continued)

- $\bar{W}_Q$ is the average queueing time for those who do queue
- $\bar{W}_{\sim Q}$ is the average queueing time for those who don't queue.
- Since $\bar{W}_{\sim Q} = 0$ the second term in this sum is zero we can rewrite in terms of $\bar{W}_Q$ as

$$\bar{W}_Q = \frac{\bar{W}}{C(m,a)} = \frac{1}{m\mu(1-\rho)} = \frac{1}{\mu(m-a)}$$

so the average queueing time for those who do queue is independent of the probability of queueing.

Example

- Example 4 at the end of the chapter
- So what is the average waiting time for the telephone service centre example? What is the queueing time for those who have to queue?

Thanks!

- company: Trent AI
- book: The Atomic Human
- twitter: @lawrennd
- The Atomic Human
- newspaper: Guardian Profile Page
- blog: http://inverseprobability.com

Further Reading

- The $M/M/m$ delay system and Erlang C of Bertsekas and Gallager (1992)

Exercises

- Tutorial Sheet 3: call-centre Erlang C of Lawrence (2006)

References

Bertsekas, D.P., Gallager, R.G., 1992. Data networks, Second. ed. Prentice-Hall, Upper Saddle River, NJ.

Lawrence, N.D., 2006. Erlang delay and loss and the M/G/1 queue.